

# Oscar Giraldo Castillo

Senior Full-Stack Engineer

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## Summary

Senior full-stack engineer with 14+ years shipping production backends and AI-native systems across fintech, edtech, agtech, medtech, and SaaS. Works end-to-end, from Python / Django, FastAPI, and Next.js services on AWS, to Rust APIs, LLM tooling (MCP, RAG, multi-provider LLMs), event-driven architectures, and ESP32 / ESP-IDF embedded firmware. Strong in distributed systems, observability and product ownership. Previously co-founded Chaty and led engineering at Artificio.

## Experience

### Senior Fullstack Developer

Oct 2024 - Present

Cometa

Remote (Ciudad de México, México)

- Member of Cometa's **finance engineering team**. Own payins, payouts, and billing / invoicing flows end-to-end across a Django modular monolith and Next.js frontend.
- Built **Argus**, a Rust read-only Redshift proxy (Axum, lambda\_http on AWS Lambda + API Gateway) powering the Guardian Credit Scoring flow: whitelisted tables, identity-based row filtering, pagination, and thread-safe credential refresh.
- Delivered the billing + inventory domain: cart payment lifecycle state machine, batch-only inventory reservation pipeline, cart-expiration via cross-module outbox events, Gandalf-authenticated endpoints, and Postgres indexes on hot paths.
- Shipped Guardian Stripe checkout end-to-end (PaymentIntent, per-currency minimum-amount drawer, locale-aware errors, ally) across Django backend and Next.js frontend.
- Build MCP servers exposing Cometa's domain APIs as LLM-callable tools; ship full-stack features on a Django modular monolith + Next.js (Clean Architecture, DDD) with on-call triage, Alembic migrations, and report-query tuning.

### Senior Software Engineer

Sep 2023 - Oct 2024

Foris

Remote

- Built airflow DAGs and create our own DSL language to create data pipelines with AWS glue jobs, S3 and Athena.
- Shipped Python backend services for academic planning and course scheduling products; led technical design reviews and mentored mid-level engineers.

### Director of Software Engineering

Jan 2023 - Sep 2023

Artificio

Remote

- Shipped an end-to-end image-search service for **Seezlo** (artwork discovery): deployed vision models on AWS SageMaker and an image-embedding RAG for artwork retrieval, exposed via FastAPI.
- Led the engineering function end-to-end: hiring, roadmap, and architecture direction for Python services on AWS.

### Python Backend Engineer

Oct 2021 - Jan 2023

SpaceAG Global

Lima, Perú

- Implemented a multi-tenant reporting module syncing AWS RDS with AWS Athena via EventBridge, Celery, and Pandas, producing Parquet files on S3.
- Scaled the Postgres master on Aurora Serverless v2 with read replicas; built AWS Lambda health-checks wired into Slack, Sentry, and Google Sheets.
- Shipped Django/Flask APIs on AWS Lambda (Serverless, Zappa); exposed a legacy Microsoft Access DB through API Gateway + DynamoDB and integrated weather stations (WeatherLink v2, Metos 2.0, Adcon).

### Backend Developer & DevOps

Jan 2020 - Dec 2021

Alfi

Remote

- Rebuilt the core application from scratch in Django and modeled all REST APIs; production stack on AWS ECS/Fargate, RDS, ElastiCache (Redis), and RabbitMQ.

- Led the backend team: story points, code reviews, and PRs.

## Co-founder & Backend Engineer

Jan 2017 - Jan 2020

*Chaty*

*Lima, Perú*

- Co-founded a multi-tenant SaaS for restaurant-chain chat ordering; integrated Facebook Messenger, WhatsApp, and Telegram APIs for chatbot-driven order flows.
- Integrated multiple payment gateways for end-to-end checkout and automated shipping assignment + order routing on PostGIS.

## Software Architect

Oct 2014 - Jan 2017

*Ventorystack*

*Remote*

- Designed and implemented a real-time inventory system on Tornado, MongoDB, AWS Lambda, and AWS SQS.

## Earlier roles: Django edtech, embedded systems, ML research

2011 - 2015

*Crehana (CTO), DGNest (Founder), Engitronic, CTIC-UNI*

*Lima, Perú*

## Projects

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### MCP Servers for Cometa | *Python, Model Context Protocol, Claude, OpenAI*

- Open-source and internal MCP server implementations exposing Cometa domain APIs (payments, admissions, communications) as LLM-callable tools for AI developer tooling.

### Chapatuplaza MCQ generator | *Python, LangChain, Gemini 2.5, Claude, Qdrant, AWS Lambda, SQS*

- Event-driven RAG for Peruvian medical-exam prep (SERUMS/MINSA): vision-based PDF OCR and image-embedding retrieval on Qdrant; multi-provider LLM generation (Gemini with Claude fallback) on AWS Lambda, with async SQS callbacks (DLQ-backed) for persistence and Langfuse observability.

### Cometag NFC reader firmware | *C, ESP-IDF, XIAO ESP32-S3, PN532, I2C*

- Embedded firmware for a XIAO ESP32-S3 driving a PN532 NFC reader over I2C; detects ISO14443A tags (MIFARE, NTAG) and streams UIDs over serial USB to prototype NFC workflows at Cometa.

### Fabrimetal | *Next.js, DRF, PostgreSQL, Docker, AWS*

- Full-stack e-commerce for laser-cutting and metal-bending services: Next.js 15 storefront, DRF backend with clean-architecture apps (controllers / services / repositories), explicit order state machine (draft, to-quote, pending-payment, paid, in-progress, delivered), Docker Compose, S3 storage, and Telegram-bot notifications for order workflow.

## Skills

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**Languages:** Python, Go, Rust, JavaScript/TypeScript, SQL, C/C++

**Backend frameworks:** Django, FastAPI, Flask, Django Ninja, Tornado, Axum (Rust)

**Frontend:** Next.js, React, TypeScript, Tailwind CSS

**Databases:** PostgreSQL, DynamoDB, MongoDB, AWS Athena, AWS Redshift, Qdrant, Pinecone

**Messaging & async:** Redis, RabbitMQ, AWS SQS, Celery, EventBridge, outbox pattern

**Cloud & DevOps:** AWS (Lambda, ECS, Fargate, S3, API Gateway, RDS, SageMaker), Docker, Kubernetes, Terraform, Serverless, CI/CD, Nginx

**AI & ML:** Model Context Protocol (MCP), LangChain, RAG, vector search, LLM providers (Claude, OpenAI, Gemini), AWS SageMaker, image embeddings, Pandas, TensorFlow

**Embedded & IoT:** ESP32 (ESP-IDF), Seeed XIAO ESP32-S3, C firmware, PN532 NFC, I2C / SPI / UART, ISO14443A (MIFARE, NTAG), microcontrollers

**Architecture & practices:** Microservices, event-driven architecture, REST, GraphQL, Clean Architecture, DDD, BDD, TDD, SCRUM, Kanban

**Observability:** OpenTelemetry (OTEL Collector), Grafana, Loki, Tempo, Prometheus, Sentry, Langfuse

**Operating systems:** GNU/Linux (primary), macOS, Windows

## Education

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### B.S. Mechatronics Engineering

2008 - 2013

*Universidad Nacional de Ingeniería (UNI)*

*Lima, Perú*

Activities: GISCIA, CEDIM, CTIC, python-uni. Relevant coursework: Matlab, algorithms.

### Marketing & Value Management (Summer course)

2016

*State University of New York at New Paltz*

*New Paltz, NY, USA*

## Certifications

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- **Curso de Go Intermedio: POO y Concurrencia**, Platzi (Dec 2025). [platzi.com/p/oscardigcast/curso/2357-golang-intermedio/diploma](https://platzi.com/p/oscardigcast/curso/2357-golang-intermedio/diploma)
- **Diploma de Especialización en IA, ML & Deep Learning con Python**, Universidad Nacional de Ingeniería, FIQT (Nov 2025), 120 hours, average grade 19/20.
- **Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning**, DeepLearning.AI (Jun 2024). [coursera.org/verify/ZZMM346D2N7T](https://coursera.org/verify/ZZMM346D2N7T)
- **AWS Certified Cloud Practitioner**, Amazon Web Services (Mar 2024). [credly.com/badges/9b39b6f3-2886-4465-b99c-b06017b88cde](https://credly.com/badges/9b39b6f3-2886-4465-b99c-b06017b88cde)

## Publications

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- **Development of a Web Database System for a Comprehensive Hypertension Management Program in the Primary Health Care Setting of Urban Areas of Peru.** *Healthcare Informatics Research* (Jan 2019), DOI 10.4258/hir.2019.25.1.41

## Honors & Awards

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- **Academic Awards:** IBS Brasil/SUNY New Paltz Executive Management (Nov 2016); UNI upper-third graduation.
- **2nd Place, Mechatronics Academic Olympics:** FESTO, Aug 2011 (pneumatic/PLC tests).

## Languages

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- **Spanish:** Native.
- **English:** Full Professional Proficiency; EF SET C2 Proficient (72/100). [cert.efset.org/en/1xUWsa](https://cert.efset.org/en/1xUWsa)